

PAYMENT GATEWAY, 3-DAY PLAN (Razorpay test mode)

E-Learning Platform. Day 1: keys and a working test payment. Day 2: the course owner's share. Day 3: invoices. Each day has a **backend** and a **frontend** PDF, and ends with a checkpoint that must work before you start the next day.

0. Honest status (read once)

- **Everything uses Razorpay TEST mode.** No real money is charged, anywhere in this plan.
- The **Java logic** (money arithmetic, earnings ledger, payouts, invoices, PDF writer, the pay-and-verify flow) ran as real Java against in-memory stand-ins: **136 checks passed, 0 failed**, plus 13 + 27 checks for the Day 1 signature and payment logic.
- The **Angular screens** ran in a real browser against a pretend backend and a stand-in for Razorpay's pop-up: **39 checks passed, 0 failed, 0 page errors** (payment, invoice page, real PDF download, My Invoices, instructor earnings, admin payout). I also built the app in the exact Day 2 state.
- **Never run here:** the Spring Boot backend wired to a real MySQL (Maven is blocked in my workspace), and a **real Razorpay payment** (it needs your keys). So expect to fix a small compile or wiring error on first start; send me the red text and I will fix it.
- Rotate any secret that was ever pasted into a chat or committed to Git.

1. What you will have on Day 3 evening

```
Customer pays (test card)
|
v
payments row PAID ---> order saved, cart emptied (Day 1)
|
+---> course_earnings: instructor's share = price - commission (Day 2)
|       |--> instructor sees it under Earnings (Waiting)
|       |--> admin sees "owed" under Payouts, pays by bank/UPI,
|               types the reference, presses Mark as paid -> instructor sees Paid
|
+---> invoices: INV-2026-000001, snapshot of the sale (Day 3)
|       |--> customer opens it on screen, downloads the PDF
```

2. Where the money really goes (important for your demo and for questions)

All student money arrives in **one** Razorpay account: the platform's. Razorpay does not know your instructors. So:

- **What we built:** the platform records how much of each sale belongs to the instructor (price minus a commission, default 20%), shows it to him, and the **admin pays him outside the app** (bank transfer or UPI) and records the reference. This is honest bookkeeping and is how many small marketplaces start.

- **What we did not build:** Razorpay **Route**, where Razorpay itself splits the payment and sends the instructor's share straight to his account. That needs Route enabled on your Razorpay account, a registered "linked account" with KYC for every instructor, and extra API calls. Day 2's guide section 7 says how it would plug into the same ledger.
- In test mode there is no real money in either case, so for a demo the story "student pays, instructor sees the earning, admin pays him out" is complete.

If someone asks "does the instructor get the money automatically?", the true answer is: "his share is calculated and recorded automatically; the transfer is done by the admin, and the automatic split is the next step (Razorpay Route)."

3. The 7 PDFs

Day	Backend	Frontend	Done when
1 (about 4 h)	PAY-DAY1-KEYS-AND-CHECKOUT-BACKEND.pdf	PAY-DAY1-KEYS-AND-CHECKOUT-FRONTEND.pdf	a test card payment enrolls the customer; a PAID row exists
2 (about 4 h)	PAY-DAY2-OWNER-EARNINGS-BACKEND.pdf	PAY-DAY2-OWNER-EARNINGS-FRONTEND.pdf	an instructor's course sale shows under his Earnings ; the admin records a payout
3 (about 4 h)	PAY-DAY3-INVOICE-BACKEND.pdf	PAY-DAY3-INVOICE-FRONTEND.pdf	the customer downloads INV-2026-000001 as a PDF

If you already did the main plan's "Day 3 afternoon: Razorpay", that is this plan's Day 1: skip to Day 2.

Two people? Day 1 and the rest: Person A does the backend guide, Person B the frontend guide of the same day, then you test together. Always finish a day's backend first, because the frontend needs its endpoints.

4. Day by day

Day 1: keys and checkout 1. Open the backend PDF, section 2: create the Razorpay account and generate **test** keys (click by click). 2. Put the keys in environment variables (section 3a), never in a file that goes to Git. 3. Copy the files, start the backend, run the 2-minute key check (section 5a). 4. Frontend PDF: switch razorpayEnabled on, pay with a test card. 5. **Checkpoint 1:** the course appears in My Enrollments and `SELECT * FROM payments` shows PAID.

Day 2: the course owner's share 1. Read the backend PDF section 0 (where the money goes) and section 1 (ledger, paise, commission). 2. Add the 8 new files, replace PaymentService, add commission.percent. 3. Frontend: the two screens, the module, the two menu links. 4. **Checkpoint 2:** an instructor-made course is bought; the instructor sees 80% of the price as Waiting; the admin marks it paid with a reference; the instructor sees Paid.

Day 3: invoices 1. Backend: 5 new files, replace PaymentRepo and PaymentService, the 5 invoice.* lines, one SecurityConfig line. 2. Frontend: 2 screens, 1 service, replace the model and the module, 4 small edits. 3. Put your real seller name, address and GSTIN (only if your company has one) into the properties. 4. **Checkpoint 3:** pay, land on the invoice, download and open the PDF.

5. All the configuration in one place (at the very end of

`application.properties`)

```
# Day 1
razorpay.key-id=${RAZORPAY_KEY_ID:}
razorpay.key-secret=${RAZORPAY_KEY_SECRET:}
razorpay.enforce=${RAZORPAY_ENFORCE:false}
# Day 2
commission.percent=${COMMISSION_PERCENT:20}
# Day 3
invoice.prefix=INV
invoice.seller-name=${INVOICE_SELLER_NAME:Online E-Learning Platform}
invoice.seller-address=${INVOICE_SELLER_ADDRESS:}
invoice.seller-gstin=${INVOICE_SELLER_GSTIN:}
invoice.gst-percent=${INVOICE_GST_PERCENT:0}
```

New tables (created by `ddl-auto=update`, no SQL to run): `payments` (Day 1), `course_earnings` and `payouts` (Day 2), `invoices` (Day 3). No existing table or endpoint changes, so the SRS and hidden tests are unaffected.

New endpoints

Method and path	Who	Day
POST <code>/api/payment/razorpay/order</code> , POST <code>/api/payment/razorpay/verify</code>	CUSTOMER	1
GET <code>/api/instructor/earnings</code>	INSTRUCTOR	2
GET <code>/api/admin/payouts/payable</code> , GET <code>/api/admin/payouts</code> , POST <code>/api/admin/payouts</code>	ADMIN	2
GET <code>/api/invoice/my</code> , <code>/by-order/{orderId}</code> , <code>/id</code> , <code>/id/pdf</code>	CUSTOMER (own), ADMIN (all)	3

Security rules. The final `SecurityConfig` in the main plan (section 8, regenerated with this plan) already contains all of them: `/api/payment/**` CUSTOMER, `/api/instructor/**` INSTRUCTOR, `/api/admin/**` ADMIN, and the one new line from Day 3: `/api/invoice/**` CUSTOMER or ADMIN.

6. Razorpay test details (for the demo)

- Card (Mastercard): **5267 3181 8797 5449**, or Visa **4111 1111 1111 1111**; any future expiry (for example 12/30); any 3-digit CVV; any name. The mock bank page then offers **Success** and **Failure**.
- UPI: type `success@razorpay` (succeeds) or `failure@razorpay` (fails).
- Razorpay can change these. If a number is refused, search "Razorpay test card details" on Razorpay's own site and use what it lists.

7. The full demo run (5 minutes)

- Admin approved an instructor earlier; the instructor adds a course "Java Basics" at Rs 500 (he sees **My courses**).
- A customer buys it with the test card, with `RAZORPAY_ENFORCE=true` so there is no other way to enroll. Show that closing the pop-up enrolls nobody.

3. The pop-up closes, the **invoice page** opens: show the number and the total, press **Download PDF**, open the file.
4. Log in as the instructor: **Earnings** shows Rs 400.00 as Waiting (500 less 20%).
5. Log in as admin: **Payouts** shows the instructor owed Rs 400.00; type `UPI - 9981`, Mark as paid.
6. Back as the instructor: Paid Rs 400.00, with the reference.

8. Going live later (not part of this plan)

- Razorpay **KYC and activation** for your business, then **Live** keys. The code does not change; only the two environment variables do.
- Add the Razorpay **webhook** (`payment.captured`) so a payment is never lost if the browser closes early.
- Decide **refunds** (and how they reverse an instructor's earning).
- Decide **GST** with your company's finance person, and fill the seller block.
- For automatic instructor payments: Razorpay **Route** (split at payment time) or RazorpayX payouts, each with their own approvals.

9. Common problems across the three days

Symptom	Likely cause	Where to look
"Razorpay is not set up on this server"	keys did not reach the backend	Day 1 backend, section 3a and 5a
502 when starting a payment	wrong key pair (Key Id from one account, Secret from another), or no internet	backend log shows Razorpay's reason
Payment works, but nothing under instructor Earnings	the course was made by an admin (no owner), or the log says "earnings could not be recorded"	Day 2 backend, section 5
Invoice page says "no invoice" after a payment	order was not paid through Razorpay, or the backend is still the Day 2 version	Day 3 backend, section 5
Start-up "No qualifying bean"	a file is missing or <code>PaymentService</code> is the wrong day's version	the day's "What goes where" table
Menu link opens the home page	<code>BillingModule</code> is listed after <code>AppRoutingModule</code>	Day 2 frontend, edit 2

10. What is not built (honest list)

Razorpay Route (automatic split), refunds and credit notes, webhook, e-mailing invoices, non-Latin characters and the rupee sign in the PDF (the on-screen invoice is fine), full GST rules (HSN/SAC, CGST/SGST/IGST), instructor bank-detail storage, tax deducted on instructor payouts. Tell me which one you want next.